

# BHARAT MISHRA

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## EDUCATION

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|-----------------------------------------------------------------------------------------------------------|------|
| <b>Bachelor of Technology</b> , VIT Bhopal University, Computer Science and Engineering <b>CGPA: 9.11</b> | 2027 |
| <b>Grade XII</b> , Vidyatree Modern World School <b>Percentage: 95.25</b>                                 | 2023 |

## TECHNICAL SKILLS

- **Programming Languages:** C/C++(Data Structure and Algorithms), Python(AI development)
- **ML/AI:** TensorFlow, Scikit-learn, Pandas, FastAPI, NumPy, OpenAI Gym, OpenCV, MediaPipe, Pydantic
- **Developer tools:** GitHub, Git, Docker, MySQL
- **Core concepts:** Deep learning, Embeddings, Reinforcement Learning, LLMs, Agentic AI, Computer Vision

## PROJECTS

**DepoIndex: AI Legal Document Indexing** *LLM APIs, Prompt Engineering, PDF Processing*  
[github.com/Bharat269/DepoIndex](https://github.com/Bharat269/DepoIndex)

- Engineered an AI-powered tool to automate the generation of a table of contents from legal deposition PDFs using an efficient preprocessing pipeline that annotates text with precise page and line numbers, **minimizing prompt token** count while enabling the LLM to extract topics with accurate source references.
- Achieved **95% page-level and 85% line-level** accuracy in topic referencing, delivering a highly reliable tool for rapid automated document indexing.

**Attentia: Adaptive Learning for Neurodivergent Children** *FastAPI, OpenCV, MediaPipe, Q-learning*  
[github.com/Bharat269/attentia.ai](https://github.com/Bharat269/attentia.ai)

- Designed an adaptive learning system that detects attention/emotional state and performs tailored interventions (e.g., adjusting course difficulty, soothing with music, reducing distraction via animations). Built the training environment for the Q-learning engine using a mix of deterministic and stochastic rules to simulate user behaviour.
- Engineered **multi-modal attention detection pipelines** (facial landmarks, head pose, audio signals) with personalized baseline calibration, using a persistent singleton architecture for efficient and accurate real-time engagement tracking and low-latency FastAPI integration.

**MovieLens: Semantic Movie Search** *FastAPI, Docker, TensorFlow, Embeddings, Recommendation Systems*  
[github.com/Bharat269/MovieMix](https://github.com/Bharat269/MovieMix)

- Deployed a production-grade semantic search system with a **under 50ms** (improved from 2 seconds) FastAPI+Docker inference pipeline to convert raw user input tags into 100-D embeddings, and retrieve movies via vectorized cosine similarity against movie embeddings derived from **matrix factorization**.
- Engineered a 500-token tag vocabulary and a hypersphere-normalized text-to-vector neural network for **solving the hubness problem** and also solving the cold-start problem by generating embeddings from tags alone achieving **0.52 cosine similarity**.

**Rescue: Environment Navigation** *Multi-Agent Reinforcement Learning, Custom Environment*  
[github.com/Bharat269/ProjectExhibition-2-Search-Rescue](https://github.com/Bharat269/ProjectExhibition-2-Search-Rescue)

- Developed autonomous agents using epsilon-greedy Q-learning within a custom-built multi-agent physics environment, optimizing pathfinding in dynamic, natural disaster simulations and **reducing search time by 90%** compared to a random-walk baseline. Simulated human behaviour with Social Forces Movement theory.

## ACHIEVEMENTS AND CERTIFICATIONS

- **Generative AI training by Kaggle and Google (2024):** Transformers, LLMs, and prompt engineering.
- **Global Rank 8261, TCS CodeVita (2024):** Solved advanced algorithmic problems.